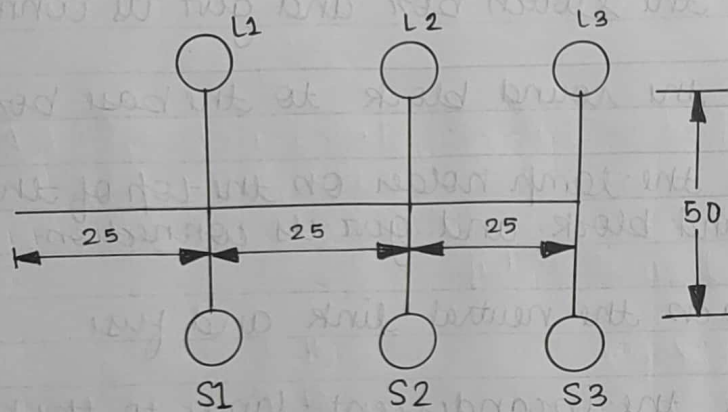
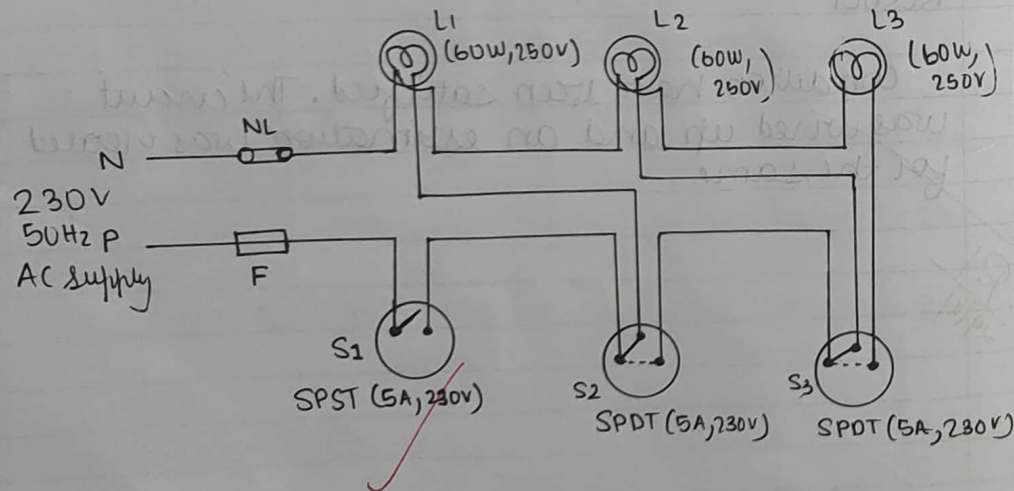


LAYOUT DIAGRAM



CIRCUIT DIAGRAM



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GODOWN WIRING

AIM

Wire up a circuit and prepare estimation for godown consist of three lamps in open conduit wiring system as per given layout.

DESIGN

$$\text{Power, } P = VI \cos \phi$$

$$I = P/V$$

$$P = 60W$$

$$V = 230V$$

$$I = \frac{60}{230} = 0.26A$$

$$\text{Load Current} = \frac{P}{230} = \underline{\underline{0.26A}}$$

Fuse Wire Rating

$$= \text{Load Current} \times 1.5A$$

$$= 0.26A \times 1.5$$

$$= \underline{\underline{0.39A}}$$

OBSERVATION TABLE

S1	S2	S3	L1	L2	L3
OFF	C → 1	C → 1	Dark	Dark	Dark
ON	C → 1	C → 1	Bright	Dark	Dark
ON	C → 2	C → 1	Dark	Bright	Dark
ON	C → 2	C → 2	Dark	Dark	Bright

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MATERIALS REQUIRED

S.No	Items	Specification	Rating	Quantity
1)	Switch	SPST	5A 230V	1
		SPDT	5A 230V	2
2)	Lamp	Incandescent	60W 250V	3
3)	Conduit	20mm PVC		225cm
4)	Copper Wire	1 sq mm		660cm
5)	Lamp holder	Straight bottom	5A	3
6)	Junction	20mm 3way PVC		1
		20mm 4way PVC		2
7)	Round Block	3inch PVC		3
8)	Switch box	3x2" PVC		3
9)	Clamp / Saddle	20mm metal		9
10)	Fuse Unit	Kit Kat (Ordinary Reminab4)	16 A	1

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11)	Fuse Wire	Tin & Lead Alloy	2A	10cm
12)	Neutral Link	Porcelene base with copper segment	16A	1
13)	Insulation Tape			1
14)	Screw	1/2"		18
		3/4"		12
		1 1/4"		3

TOOLS REQUIRED

Screw driver, line-tester, combination plier, ball-peen hammer, poker, nose plier, wire strippers.

PROCEDURE

1. Draw the circuit diagram and check whether it is correct
2. Fix the conduit to the board using saddle of appropriate diameter
3. Wire the circuit as per the circuit diagram.

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4. Fix the switch box and give its connections.
5. Fix the round block to the base board.
6. Fix the lamp holder on the top of the round block and give its connections.
7. Attach the neutral link and fuse.
8. Add the incandescent lamp to the holder and turn the switch on.
9. Record the observation on the observation table.

RESULT

wired up the circuit and prepared estimation for the godown wiring configuration.

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